=== UNIT 5: WAVES, ACOUSTICS & INDUSTRIAL ULTRASONICS ===

>> 5.1 Simple Harmonic Motion and Mechanical Waves

Wave mechanics governs the transport of mechanical disturbance and energy through deformable elastic media without net translation of matter.
* Simple Harmonic Motion (SHM): Restoring force is directly proportional to displacement from equilibrium: F = - k * x => d^2x/dt^2 + omega_0^2 * x = 0, where natural angular frequency omega_0 = sqrt(k / m).
* One-Dimensional Wave Equation: A spatial disturbance y(x,t) propagating with wave velocity v satisfies: d^2y/dx^2 = (1 / v^2) * (d^2y/dt^2).
* Wave Parameters: Wavelength lambda, Frequency f = omega / (2*pi), Wave velocity v = f * lambda, Wave number k = 2*pi / lambda.

>> 5.2 Damped and Forced Oscillations & Resonance

In physical systems, dissipative frictional forces (viscous damping proportional to velocity F_d = - b * dx/dt) retard motion:
* Differential Equation of Damped Harmonic Oscillator: m * (d^2x/dt^2) + b * (dx/dt) + k * x = 0 => d^2x/dt^2 + 2*gamma*(dx/dt) + omega_0^2 * x = 0, where gamma = b / (2*m) is damping factor.
* Damping Regimes:
- 1. Underdamped (gamma < omega_0): Oscillatory decay with reduced frequency omega_d = sqrt(omega_0^2 - gamma^2). Amplitude decays exponentially: A(t) = A_0 * e^(- gamma * t).
- 2. Critically Damped (gamma = omega_0): System returns to equilibrium in minimum possible time without oscillation (ideal for civil seismographs and vehicle suspension dampers).
- 3. Overdamped (gamma > omega_0): Heavy non-oscillatory creeping return to equilibrium.
* Logarithmic Decrement (delta): Natural log of ratio of two consecutive peak amplitudes: delta = ln(A_n / A_{n+1}) = gamma * T_d = 2 * pi * gamma / omega_d.
* Quality Factor (Q): Dimensionless parameter measuring sharpness of resonance: Q = 2*pi * (Stored Energy / Energy Lost per Cycle) = omega_0 / (2*gamma) = pi / delta.
* Forced Oscillations and Amplitude Resonance: Under external driving force F0*cos(omega*t), maximum displacement amplitude occurs when driving frequency matches natural frequency: omega_res = sqrt(omega_0^2 - 2*gamma^2).

>> 5.3 Fundamentals and Properties of Ultrasonic Waves

Ultrasonic waves are acoustic longitudinal mechanical waves having frequencies strictly above the upper audible human limit (f > 20 kHz):
* Distinctive Properties: (1) Extremely short acoustic wavelengths (sub-millimeter in solids), (2) Highly directional beam propagation with negligible diffraction

spreading, (3) High energy density, (4) Capable of inducing acoustic cavitation in liquids, (5) Highly sensitive to acoustic impedance mismatches at material interfaces.
* Characteristic Acoustic Impedance (Z): Measures resistance of a medium to ultrasonic wave transmission: Z = rho * v, where rho is density and v is acoustic velocity in medium (Rayls or kg/m^2-s).
* Acoustic Reflection and Transmission at Boundaries:
- Sound Pressure Reflection Coefficient: R_p = (Z2 - Z1) / (Z2 + Z1).
- Sound Pressure Transmission Coefficient: T_p = (2 * Z1) / (Z2 + Z1).
- If Z1 matches Z2 (acoustic matching), 100% of ultrasonic energy transmits across the interface.

>> 5.4 Production Methods for Ultrasonics

* 1. Magnetostriction Method (Joule Effect):
- Principle: When a ferromagnetic rod (iron or nickel) is magnetized by an alternating magnetic field, it undergoes mechanical elastic deformation (dimensional length change Delta L) irrespective of field polarity.
- Construction: Nickel rod clamped at center inside an inductive coil connected to a high-frequency LC tank oscillator.
- Resonant Frequency: f = (p / (2 * L)) * sqrt(Y / rho), where p = 1,2,3... harmonic, Y is Young’s modulus, L is rod length.
- Limitation: Usable only for lower ultrasonic frequencies (up to ~100 kHz) due to heavy eddy current and hysteresis losses at higher frequencies.
* 2. Piezoelectric Oscillator Method (Curie Brothers’ Effect):
- Principle: Inverse Piezoelectric Effect. Applying an alternating RF electric field across opposite faces of certain non-centrosymmetric crystals (Quartz, PZT - Lead Zirconate Titanate) causes mechanical vibrations (compression and rarefaction) at identical frequency.
- Resonant Condition: Occurs when applied RF electric frequency matches fundamental mechanical resonant frequency of the crystal slice:
Formula: f = (p / (2 * t)) * sqrt(Y / rho), where t is crystal plate thickness.
- Advantages: Operates up to ultra-high frequencies (hundreds of MHz), highly stable, compact.

>> 5.5 Detection of Ultrasonic Waves

* 1. Piezoelectric Receiver: Direct piezoelectric effect generates electric potential across crystal faces when ultrasonic waves strike them.
* 2. Acoustic Diffraction Grating Method: Ultrasonic longitudinal waves set up periodic spatial compressions and rarefactions in a transparent liquid, modulating its local refractive index to act as an optical diffraction grating for monochromatic light.
* 3. Kundt’s Tube with Lycopodium Powder: Nodal and antinodal stationary wave

patterns produce distinct striations.
* 4. Thermal Detector: Fine platinum wire resistance shifts due to adiabatic heating at wave antinodes.

>> 5.6 Civil Engineering & Industrial Applications of Ultrasonics

* 1. Non-Destructive Testing (NDT) of Concrete Structures: Ultrasonic Pulse Velocity (UPV) testing measures transit time of longitudinal ultrasound through structural concrete columns, slabs, and bridge decks to determine in-situ compressive strength, detect internal honeycomb voids, and assess fire damage.
* 2. Flaw Detection in Steel Welds and Bridges: Pulse-echo ultrasonic testing detects internal cracks, slag inclusions, and fatigue fissures in structural steel joints.
* 3. SONAR and Bathymetry: Sound Navigation and Ranging maps riverbeds, sea floor topography, and scours around bridge foundation piers.
* 4. Ultrasonic Drilling, Cutting, and Welding: Micro-drilling of hard brittle civil stones (granite, quartz, glass).

[EXAMPLE 1: Resonant Frequency of Piezoelectric Quartz Plate]

Problem Context: An X-cut quartz crystal plate of thickness t = 2.0 mm is used in an ultrasonic generator. Properties of quartz are Young's modulus Y = 7.90 * 10^10 N/m^2 and density rho = 2650 kg/m^3. Calculate: (1) Fundamental resonant frequency f1, (2) Acoustic wavelength in quartz, (3) Thickness required to generate ultrasound at f = 5.0 MHz.
Solution Step 1: Acoustic Velocity in Quartz v:
v = sqrt(Y / rho) = sqrt(7.90 * 10^10 / 2650) = sqrt(29,811,320.75) = 5460.0 m/s.
Solution Step 2: Fundamental Resonant Frequency f1 (p = 1):
f1 = v / (2 * t) = 5460.0 / (2 * 0.002 m) = 5460.0 / 0.004 = 1,365,000 Hz = 1.365 MHz.
Solution Step 3: Acoustic Wavelength lambda in Quartz:
lambda = v / f1 = 5460.0 / 1.365 * 10^6 = 0.004 m = 4.0 mm (= 2 * t).
Solution Step 4: Thickness for f = 5.0 MHz:
t_new = v / (2 * f_new) = 5460.0 / (2 * 5.0 * 10^6) = 5460.0 / 10^7 = 5.46 * 10^-4 m = 0.546 mm.
Conclusion: Fundamental frequency = 1.365 MHz; acoustic wavelength = 4.0 mm; required thickness for 5 MHz is 0.546 mm.

[EXAMPLE 2: Damped Oscillator Logarithmic Decrement & Q-Factor]

Problem Context: An underdamped mechanical vibration sensor of mass m = 0.25 kg has spring stiffness k = 100 N/m and damping coefficient b = 0.50 N-s/m. Calculate: (1) Natural frequency omega_0, (2) Damping factor gamma and damped frequency omega_d, (3)

Logarithmic decrement delta, (4) Quality factor Q, (5) Time required for vibration amplitude to drop to 10% of initial value.
Solution Step 1: Natural Frequency omega_0:
omega_0 = sqrt(k / m) = sqrt(100 / 0.25) = sqrt(400) = 20.0 rad/s.
Solution Step 2: Damping Factor gamma and Damped Frequency omega_d:
gamma = b / (2 * m) = 0.50 / (2 * 0.25) = 1.0 s^-1.
omega_d = sqrt(omega_0^2 - gamma^2) = sqrt(20^2 - 1^2) = sqrt(399) = 19.975 rad/s.
Solution Step 3: Logarithmic Decrement delta:
Damped period T_d = 2 * pi / omega_d = 2 * 3.14159 / 19.975 = 0.3145 s.
delta = gamma * T_d = 1.0 * 0.3145 = 0.3145.
Solution Step 4: Quality Factor Q:
Q = omega_0 / (2 * gamma) = 20.0 / (2 * 1.0) = 10.0.
Solution Step 5: Time for 90% Amplitude Decay (A(t) / A_0 = 0.10):
A(t) = A_0 * e^(- gamma * t) => e^(- 1.0 * t) = 0.10.
- t = ln(0.10) = - 2.3026 => t = 2.30 seconds.
Conclusion: omega_d = 19.98 rad/s; delta = 0.315; Q = 10.0; decay time to 10% amplitude = 2.30 s.

[EXAMPLE 3: Ultrasonic Pulse Velocity (UPV) Testing of Concrete Column]

Problem Context: An Ultrasonic Pulse Velocity (UPV) test is performed across a reinforced concrete bridge pier of path length L = 450 mm (0.45 m). The measured transit time of the ultrasonic longitudinal pulse is Delta t = 105 microseconds. Standard civil engineering quality ratings (IS 13311 Part 1): v > 4.5 km/s (Excellent); 3.5 to 4.5 km/s (Good); 3.0 to 3.5 km/s (Medium); < 3.0 km/s (Doubtful). Determine: (1) Ultrasonic pulse velocity in concrete, (2) Dynamic Young’s modulus E_d if concrete density is rho = 2400 kg/m^3 and Poisson’s ratio is nu = 0.20, (3) Quality classification of the concrete column.
Solution Step 1: Compute Ultrasonic Pulse Velocity v:
v = Path Length / Transit Time = 0.450 m / (105 * 10^-6 s) = 4285.7 m/s = 4.29 km/s.
Solution Step 2: Compute Dynamic Young’s Modulus E_d:
Relation for longitudinal wave in an elastic solid:
v = sqrt([E_d * (1 - nu)] / [rho * (1 + nu) * (1 - 2*nu)]).
Rearranging: E_d = rho * v^2 * [(1 + nu) * (1 - 2*nu) / (1 - nu)].
Poisson factor = [(1 + 0.20) * (1 - 0.40)] / (1 - 0.20) = [1.20 * 0.60] / 0.80 = 0.72 / 0.80 = 0.90.
E_d = 2400 kg/m^3 * (4285.7 m/s)^2 * 0.90 = 2400 * (1.8367 * 10^7) * 0.90 = 3.967 * 10^10 N/m^2 = 39.67 GPa.
Solution Step 3: Quality Classification:
Calculated velocity v = 4.29 km/s lies between 3.5 km/s and 4.5 km/s.
Conclusion: Pulse velocity = 4.29 km/s; Dynamic modulus E_d = 39.67 GPa; Concrete rating is GOOD (satisfactory structural integrity without internal honeycombing).

[EXAMPLE 4: Acoustic Grating Measurement of Sound Velocity in Liquid]

Problem Context: In an acoustic grating experiment, an ultrasonic wave of frequency f = 3.0 MHz is set up in an organic chemical solvent. A monochromatic sodium light beam (lambda = 589.3 nm) passes perpendicularly through the liquid cell, producing diffraction fringes on a screen placed at focal distance D = 1.2 m. The distance between the central zeroth-order and first-order diffraction maxima is x_1 = 3.5 mm. Calculate: (1) Grating element d (acoustic wavelength in liquid), (2) Acoustic velocity v of ultrasound in the solvent.

Solution Step 1: Grating Equation for First-Order Diffraction:

Diffraction angle: sin(theta) ~= tan(theta) = x_1 / D = 0.0035 m / 1.2 m = 0.002917 radians.

Grating condition: d * sin(theta) = 1 * lambda => d = lambda / sin(theta).

Acoustic wavelength d = (589.3 * 10^-9 m) / 0.002917 = 2.0205 * 10^-4 m = 0.202 mm.

Solution Step 2: Compute Ultrasonic Velocity v in the liquid:

v = f * d = (3.0 * 10^6 s^-1) * (2.0205 * 10^-4 m) = 606.15 m/s.

Conclusion: Grating pitch d = 0.202 mm; Ultrasonic wave velocity in liquid = 606.15 m/s.

>> 5.7 University Review & Examination Questions

* Q1: Explain the principle, construction, and working of a Piezoelectric Oscillator for producing ultrasonic waves.

Answer: Principle: Inverse piezoelectric effect. When alternating RF electrical voltage is applied across opposite metallized faces of an anisotropic crystal plate (quartz), the crystal alternately expands and contracts along its optical/mechanical axis at the driving frequency. Construction: An X-cut quartz plate is mounted between metal plates and connected in parallel with an inductance coil L1 forming a tank circuit coupled to the grid of a vacuum tube or transistor RF oscillator. A high-voltage DC supply powers the circuit. Working: Tuning the variable capacitor C1 adjusts electrical frequency. When RF frequency coincides with the natural fundamental mechanical frequency of the crystal slice (f = (p/2t)*sqrt(Y/rho)), acoustic resonance occurs, radiating intense ultrasonic waves into the adjacent medium.

* Q2: Explain the principle of Ultrasonic Pulse Velocity (UPV) testing in concrete structures and its clinical engineering interpretation.

Answer: Principle: An electro-acoustic transducer placed on the concrete surface converts high-voltage electrical pulses into longitudinal ultrasonic pulses (~54 kHz). The pulse traverses concrete path length L to a receiving transducer, which reconverts acoustic vibrations into electrical signals displayed on a digital timer. Transit time t is recorded to compute pulse velocity v = L / t. Engineering Interpretation: Wave speed depends directly on concrete density and elastic modulus (v = sqrt(E*(1-nu) / rho*(1+nu)*(1-2nu))). Higher velocity (> 4.5 km/s) indicates dense, defect-free concrete with high compressive strength. Lower velocity (< 3.0 km/s) warns of internal honeycomb voids, micro-cracks, freeze-thaw damage, or structural delamination.

>> 5.8 Key Takeaways and Revision Points

* Underdamped oscillations decay exponentially with logarithmic decrement delta = gamma * T_d.
* Piezoelectric generators produce high-frequency ultrasound (> 1 MHz) via inverse piezoelectric effect.
* Magnetostriction generators produce lower-frequency ultrasound (< 100 kHz) via Joule effect.
* UPV test is the universal civil non-destructive standard for concrete compressive strength and void detection.